

## ARRAYS

### Aim:

To understand and implement array operations in Java.

### PRE LAB EXERCISE

#### QUESTIONS

- ✓ What is an array?

It is a collection of data items of same data type.

- ✓ Why are arrays used?

->Store multiple values under single name.

->It makes easy access and faster execution.

- ✓ What is the difference between array and variable?

| Variable                                | Array                            |
|-----------------------------------------|----------------------------------|
| Stores only one value                   | Stores multiple values           |
| Needs separate variables for each value | Uses one name for many values    |
| Example: int a = 10;                    | Example: int a[] = {10, 20, 30}; |

### IN LAB EXERCISE

#### Objective:

To perform array operations using simple programs.

#### PROGRAMS:

##### 1. Program to Read and Print Array Elements

#### Code:

```
import java.util.Scanner;
```

```
public class ReadPrintArray {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        int[] arr = new int[5];  
        System.out.println("Enter 5 elements:");  
        for(int i = 0; i < 5; i++)  
            arr[i] = sc.nextInt();  
        System.out.println("Array elements are:");  
        for(int i = 0; i < 5; i++)  
            System.out.print(arr[i] + " ");  
    }  
}
```

**OUTPUT:**

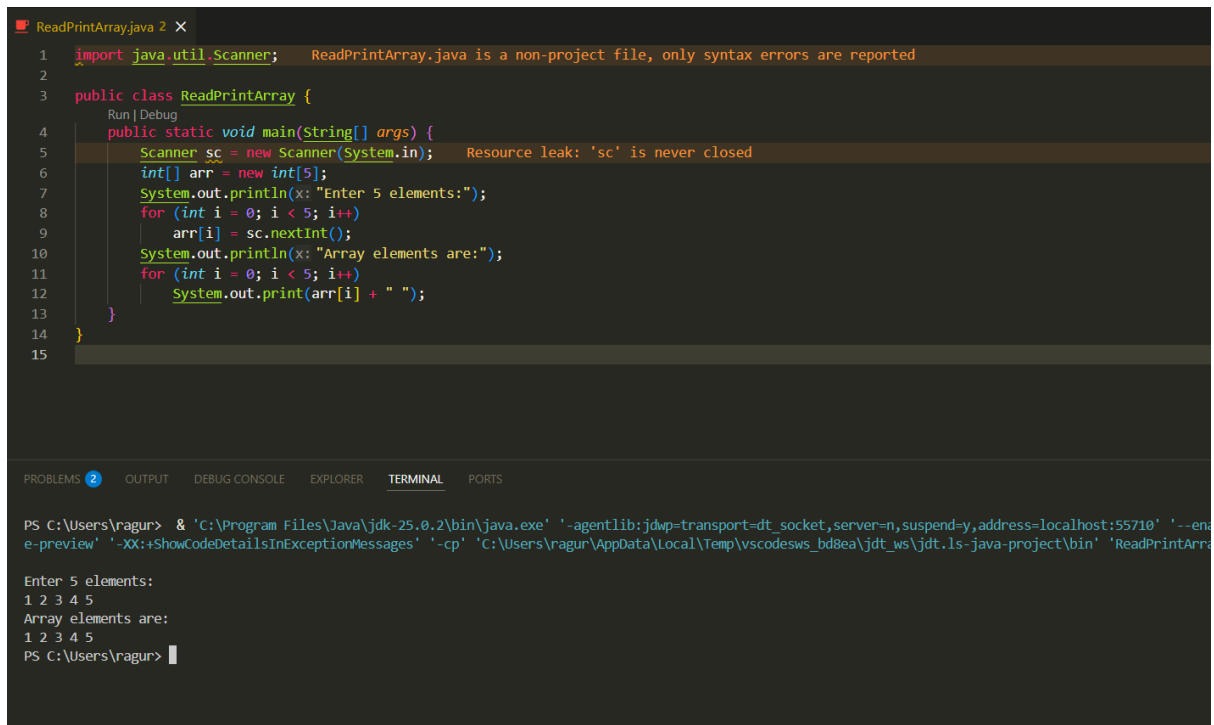
**Input:**

**10 20 30 40 50**

**Output:**

**Array elements are:**

**10 20 30 40 50**



```
1 import java.util.Scanner;
2
3 public class ReadPrintArray {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         int[] arr = new int[5];
7         System.out.println("Enter 5 elements:");
8         for (int i = 0; i < 5; i++)
9             arr[i] = sc.nextInt();
10        System.out.println("Array elements are:");
11        for (int i = 0; i < 5; i++)
12            System.out.print(arr[i] + " ");
13    }
14 }
15
```

PS C:\Users\ragur> & 'C:\Program Files\Java\jdk-25.0.2\bin\java.exe' '-agentlib:jdwp=transport=dt\_socket,server=n,suspend=y,address=localhost:55710' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\ragur\AppData\Local\Temp\vscodesws\_bd8ea\jdt\_ws\jdt.ls-java-project\bin' 'ReadPrintArray'

Enter 5 elements:  
1 2 3 4 5  
Array elements are:  
1 2 3 4 5  
PS C:\Users\ragur>

## 2. Program to Find Sum of Array Elements

Code:

```
import java.util.Scanner;

public class SumArray {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int[] arr = new int[5];

        int sum = 0;

        System.out.println("Enter 5 elements:");

        for(int i = 0; i < 5; i++)

            arr[i] = sc.nextInt();

        for(int i = 0; i < 5; i++)

            sum += arr[i];

        System.out.println("Sum = " + sum);

    }
}
```

}

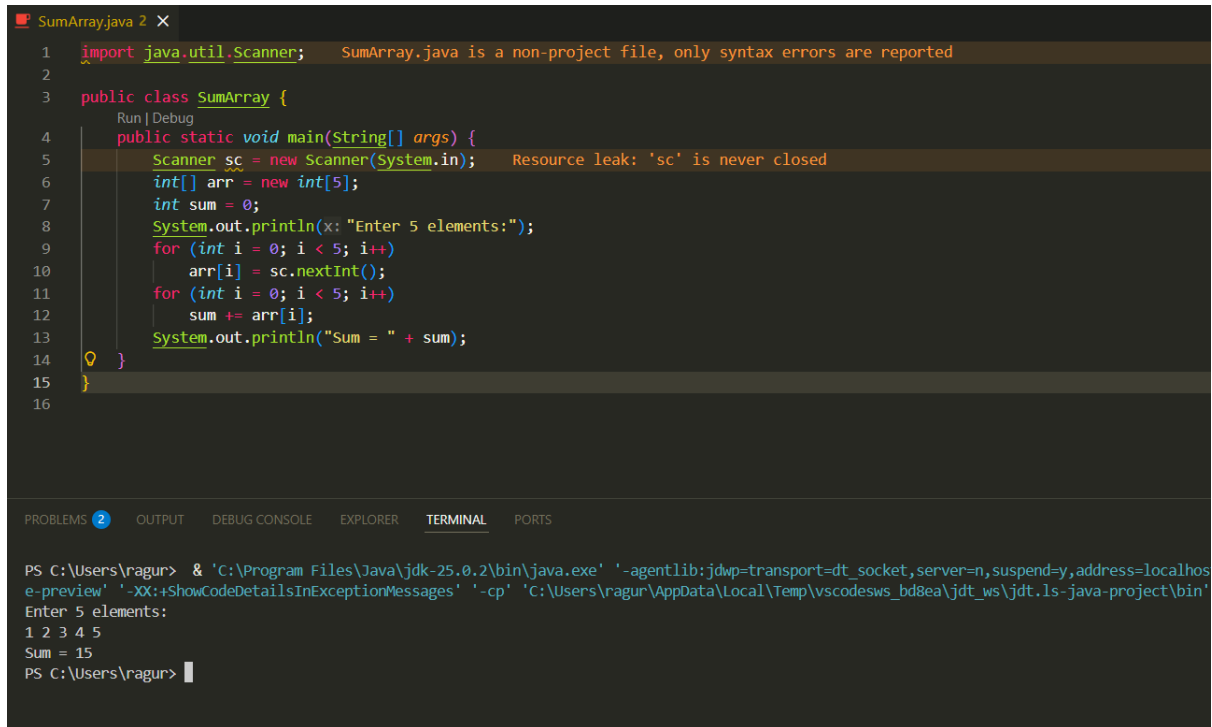
## OUTPUT:

Input:

5 10 15 20 25

Output:

Sum = 75



```
1 import java.util.Scanner;
2
3 public class SumArray {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         int[] arr = new int[5];
7         int sum = 0;
8         System.out.println("Enter 5 elements:");
9         for (int i = 0; i < 5; i++)
10             arr[i] = sc.nextInt();
11         for (int i = 0; i < 5; i++)
12             sum += arr[i];
13         System.out.println("Sum = " + sum);
14     }
15 }
16
```

PS C:\Users\ragur> & 'C:\Program Files\Java\jdk-25.0.2\bin\java.exe' '-agentlib:jdwp=transport=dt\_socket,server=n,suspend=y,address=localhos  
e-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\ragur\AppData\Local\Temp\vscode\vs\_b8ea\jdt\_ws\jdt.ls-java-project\bin'  
Enter 5 elements:  
1 2 3 4 5  
Sum = 15  
PS C:\Users\ragur>

## 3. Program to Find Largest Element in an Array

Code:

```
import java.util.Scanner;
```

```
public class LargestElement {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        int[] arr = new int[5];
```

```
        System.out.println("Enter 5 elements:");
```

```
        for(int i = 0; i < 5; i++)
```

```
            arr[i] = sc.nextInt();
```

```
        int max = arr[0];
```

```

        for(int i = 1; i < 5; i++)

            if(arr[i] > max)

                max = arr[i];

        System.out.println("Largest element = " + max);

    }

}

```

**OUTPUT:**

**Input:**

**12 45 23 9 30**

**Output:**

**Largest element = 45**

```

LargestElement.java 2 X
1  import java.util.Scanner;   LargestElement.java is a non-project file, only syntax errors are reported
2
3  public class LargestElement {
4      Run | Debug
5      public static void main(String[] args) {
6          Scanner sc = new Scanner(System.in);   Resource leak: 'sc' is never closed
7          int[] arr = new int[5];
8          System.out.println(x: "Enter 5 elements:");
9          for (int i = 0; i < 5; i++)
10             arr[i] = sc.nextInt();
11             int max = arr[0];
12             for (int i = 1; i < 5; i++)
13                 if (arr[i] > max)
14                     max = arr[i];
15             System.out.println("Largest element = " + max);
16         }
17     }

```

PROBLEMS 2 OUTPUT DEBUG CONSOLE EXPLORER TERMINAL PORTS

```

PS C:\Users\ragur> & 'C:\Program Files\Java\jdk-25.0.2\bin\java.exe' '-agentlib:jdwp=transport=dt_socket,server=n,suspend=y,address=localhost:63060'
e-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\ragur\AppData\Local\Temp\vscodesws_bd8ea\jdt_ws\jdt.ls-java-project\bin' 'Larges

```

Enter 5 elements:  
1 2 3 4 5  
Largest element = 5  
PS C:\Users\ragur> |

#### 4. Program to Reverse an Array

**Code:**

```
import java.util.Scanner;
```

```
public class ReverseArray {
```

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```

public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);

    int[] arr = new int[5];

    System.out.println("Enter 5 elements:");

    for(int i = 0; i < 5; i++)

        arr[i] = sc.nextInt();

    System.out.println("Reversed array:");

    for(int i = 4; i >= 0; i--)

        System.out.print(arr[i] + " ");

    }
}

```

**OUTPUT:**

**Input:**

**1 2 3 4 5**

**Output:**

**Reversed array:**

**5 4 3 2 1**

```

ReverseArray.java 2 X
1  import java.util.Scanner;
2
3  public class ReverseArray {
4      public static void main(String[] args) {
5          Scanner sc = new Scanner(System.in);
6          int[] arr = new int[5];
7          System.out.println("Enter 5 elements:");
8          for (int i = 0; i < 5; i++)
9              arr[i] = sc.nextInt();
10         System.out.println("Reversed array:");
11         for (int i = 4; i >= 0; i--)
12             System.out.print(arr[i] + " ");
13     }
14 }
15

```

PROBLEMS 2 OUTPUT DEBUG CONSOLE EXPLORER TERMINAL PORTS

```

PS C:\Users\ragur> & 'C:\Program Files\Java\jdk-25.0.2\bin\java.exe' '-agentlib:jdwp=transport=dt_socket,server=n,suspend=y,address=localhost:63072'
e-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\ragur\AppData\Local\Temp\vscodesws_bd8ea\jdt_ws\jdt.ls-java-project\bin' 'Revers
Enter 5 elements:
1 2 3 4 5
Reversed array:
5 4 3 2 1
PS C:\Users\ragur>

```

## 5. Program to Count Even and Odd Numbers

Code:

```
import java.util.Scanner;

public class EvenOddCount {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int[] arr = new int[5];

        int even = 0, odd = 0;

        System.out.println("Enter 5 elements:");

        for(int i = 0; i < 5; i++)

            arr[i] = sc.nextInt();

        for(int i = 0; i < 5; i++) {

            if(arr[i] % 2 == 0)

                even++;

            else

                odd++;

        }

        System.out.println("Even = " + even);

        System.out.println("Odd = " + odd);

    }

}
```

**OUTPUT:**

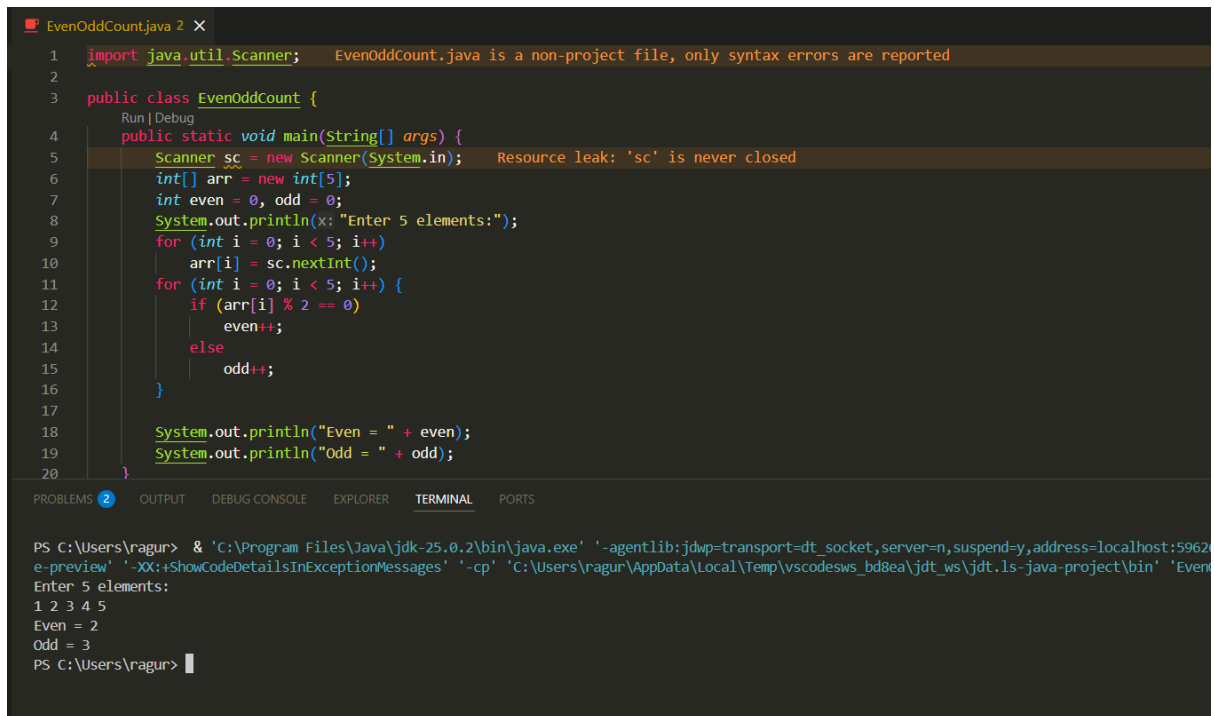
**Input:**

**2 7 4 9 10**

**Output:**

**Even = 3**

**Odd = 2**



```
1 import java.util.Scanner; EvenOddCount.java is a non-project file, only syntax errors are reported
2
3 public class EvenOddCount {
4     Run | Debug
5     public static void main(String[] args) {
6         Scanner sc = new Scanner(System.in); Resource leak: 'sc' is never closed
7         int[] arr = new int[5];
8         int even = 0, odd = 0;
9         System.out.println(x: "Enter 5 elements:");
10        for (int i = 0; i < 5; i++)
11            arr[i] = sc.nextInt();
12        for (int i = 0; i < 5; i++) {
13            if (arr[i] % 2 == 0)
14                even++;
15            else
16                odd++;
17        }
18        System.out.println("Even = " + even);
19        System.out.println("Odd = " + odd);
20    }
}
```

PROBLEMS 2 OUTPUT DEBUG CONSOLE EXPLORER TERMINAL PORTS

```
PS C:\Users\ragur> & 'C:\Program Files\Java\jdk-25.0.2\bin\java.exe' '-agentlib:jdwp=transport=dt_socket,server=n,suspend=y,address=localhost:5962
e-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\ragur\AppData\Local\Temp\vscodesws_bd8ea\jdt_ws\jdt.ls-java-project\bin' 'Even
Enter 5 elements:
1 2 3 4 5
Even = 2
Odd = 3
PS C:\Users\ragur>
```

## 6. Program to Sort Array in Ascending Order

Code:

```
import java.util.Scanner;

public class SortArray {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int[] arr = new int[5];

        int temp;

        System.out.println("Enter 5 elements:");

        for(int i = 0; i < 5; i++)

            arr[i] = sc.nextInt();

        for(int i = 0; i < 5; i++) {

            for(int j = i + 1; j < 5; j++) {

                if(arr[i] > arr[j]) {

                    temp = arr[i];

                    arr[i] = arr[j];
```



```

        arr[j] = temp;
    }
}

System.out.println("Sorted array:");

for(int i = 0; i < 5; i++)

    System.out.print(arr[i] + " ");

}
}

```

## OUTPUT:

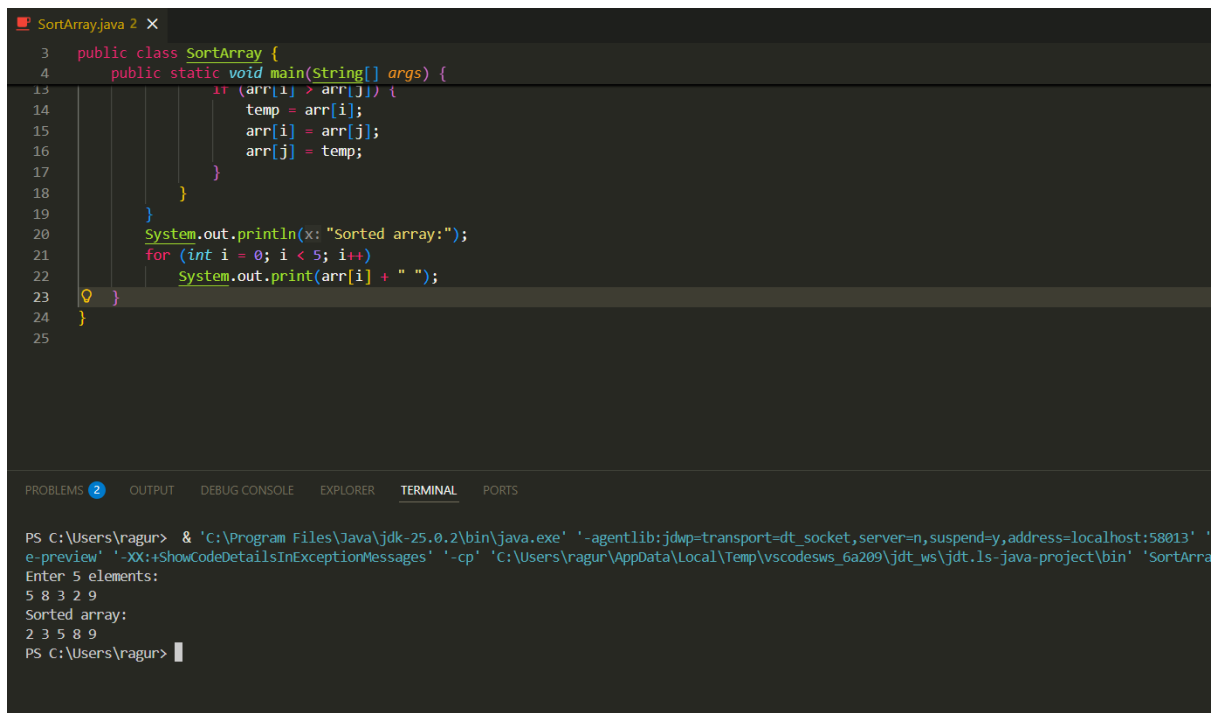
Input:

45 12 78 23 9

Output:

Sorted array:

9 12 23 45 78



The screenshot shows a Java IDE with a file named 'SortArray.java'. The code is as follows:

```

3 public class SortArray {
4     public static void main(String[] args) {
13         if (arr[i] > arr[j]) {
14             temp = arr[i];
15             arr[i] = arr[j];
16             arr[j] = temp;
17         }
18     }
19 }
20 System.out.println("Sorted array:");
21 for (int i = 0; i < 5; i++)
22     System.out.print(arr[i] + " ");
23 }
24 }
25

```

The terminal output shows the execution of the program:

```

PS C:\Users\ragur> & 'C:\Program Files\Java\jdk-25.0.2\bin\java.exe' '-agentlib:jdwp=transport=dt_socket,server=n,suspend=y,address=localhost:58013' '
e-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\ragur\AppData\Local\Temp\vscodesws_6a209\jdt_ws\jdt.ls-java-project\bin' 'SortArra
Enter 5 elements:
5 8 3 2 9
Sorted array:
2 3 5 8 9
PS C:\Users\ragur>

```

## 7. Program to Find Second Largest Element

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**Code:**

```
import java.util.Scanner;

public class SecondLargest {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int[] arr = new int[5];

        System.out.println("Enter 5 elements:");

        for(int i = 0; i < 5; i++)

            arr[i] = sc.nextInt();

        int largest = arr[0];

        int second = arr[0];

        for(int i = 0; i < 5; i++) {

            if(arr[i] > largest) {

                second = largest;

                largest = arr[i];

            }

        }

        System.out.println("Second largest = " + second);

    }

}
```

**OUTPUT:**

**Input:**

**10 45 23 89 67**

**Output:**

**Second largest = 67**

```
SecondLargest.java 2 X
3 public class SecondLargest {
4     public static void main(String[] args) {
15         second = largest;
16         largest = arr[i];
17     }
18 }
19 System.out.println("Second largest = " + second);
20 }
21 }
22 }

PROBLEMS 2 OUTPUT DEBUG CONSOLE EXPLORER TERMINAL PORTS
PS C:\Users\ragur> & 'C:\Program Files\Java\jdk-25.0.2\bin\java.exe' '-agentlib:jdwp=transport=dt_socket,server=n,suspend=y,address=localhost:6
e-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\ragur\AppData\Local\Temp\vscodesws_6a209\jdt_ws\jdt.ls-java-project\bin' 'S
Enter 5 elements:
6 9 3 4 2
Second largest = 6
PS C:\Users\ragur>
```

## 8. Program for Matrix Addition (2D Array)

Code:

```
import java.util.Scanner;
```

```
public class MatrixAddition {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        int[][] a = new int[2][2];
```

```
        int[][] b = new int[2][2];
```

```
        int[][] sum = new int[2][2];
```

```
        System.out.println("Enter elements of matrix A:");
```

```
        for(int i = 0; i < 2; i++)
```

```
            for(int j = 0; j < 2; j++)
```

```
                a[i][j] = sc.nextInt();
```

```
        System.out.println("Enter elements of matrix B:");
```

```
        for(int i = 0; i < 2; i++)
```

```

        for(int j = 0; j < 2; j++)
            b[i][j] = sc.nextInt();
    for(int i = 0; i < 2; i++)
        for(int j = 0; j < 2; j++)
            sum[i][j] = a[i][j] + b[i][j];
    System.out.println("Sum matrix:");
    for(int i = 0; i < 2; i++) {
        for(int j = 0; j < 2; j++)
            System.out.print(sum[i][j] + " ");
        System.out.println();
    }
}

```

**OUTPUT:**

**Matrix A:**

**1 2**

**3 4**

**Matrix B:**

**5 6**

**7 8**

**Sum matrix:**

**6 8**

**10 12**

```
MatrixAddition.java 2 x
1 import java.util.Scanner; MatrixAddition.java is a non-project file, only syntax errors are reported
2
3 public class MatrixAddition {
4     Run | Debug
5     public static void main(String[] args) {
6         Scanner sc = new Scanner(System.in); Resource Leak: 'sc' is never closed
7         int[][] a = new int[2][2];
8         int[][] b = new int[2][2];
9         int[][] sum = new int[2][2];
10        System.out.println("Enter elements of matrix A:");
11        for (int i = 0; i < 2; i++)
12            for (int j = 0; j < 2; j++)
13                a[i][j] = sc.nextInt();
14        System.out.println("Enter elements of matrix B:");
15        for (int i = 0; i < 2; i++)
16            for (int j = 0; j < 2; j++)
17                b[i][j] = sc.nextInt();
18        for (int i = 0; i < 2; i++)
19            for (int j = 0; j < 2; j++)
20                sum[i][j] = a[i][j] + b[i][j];
21        System.out.println("Sum matrix:");
22    }
23 }

PROBLEMS 2 OUTPUT DEBUG CONSOLE EXPLORER TERMINAL PORTS

PS C:\Users\ragur> & 'C:\Program Files\Java\jdk-25.0.2\bin\java.exe' '-agentlib:jdwp=transport=dt_socket,server=n,suspend=y,address=localhost:54299' '-.
e-preview' '-XX:ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\ragur\AppData\Local\Temp\vscodesws_6a209\jdt_ws\jdt.ls-java-project\bin' 'MatrixAdd

Enter elements of matrix A:
1 2 3 4
Enter elements of matrix B:
1 2 3 4
Sum matrix:
2 4
6 8
PS C:\Users\ragur> |
```

## POST LAB EXERCISE

- ✓ Why is array indexing usually started from zero instead of one?

Because index 0 means the first memory location, making access faster and easier.

- ✓ What happens if we try to access an array element outside its declared size?

It causes an error.

- Java → `ArrayIndexOutOfBoundsException`

- ✓ How does memory allocation differ for static arrays and dynamic arrays?

- **Static array:** Size is fixed, decided early
- **Dynamic array:** Size can change during program execution

- ✓ **Why is searching faster in arrays compared to linked lists?**

**Because arrays allow direct access using index, but linked lists do not.**

- ✓ **What is the difference between contiguous and non-contiguous memory allocation?**

- **Contiguous:** Stored together → faster (arrays)
- **Non-contiguous:** Stored separately → slower (linked lists)

**Result:**

**Thus the array operations were executed successfully.**

## **ASSESSMENT**

| <b>Description</b>       | <b>Max Marks</b> | <b>Marks Awarded</b> |
|--------------------------|------------------|----------------------|
| <b>Pre Lab Exercise</b>  | <b>5</b>         |                      |
| <b>In Lab Exercise</b>   | <b>10</b>        |                      |
| <b>Post Lab Exercise</b> | <b>5</b>         |                      |
| <b>Viva</b>              | <b>10</b>        |                      |
| <b>Total</b>             | <b>30</b>        |                      |
| <b>Faculty Signature</b> |                  |                      |

